

FALCON's Response to RFQ DHL for Supply of Conveyor Automation System

February 18, 2023



www.falconautotech.com



Kind Attention –

Mr. Nishant Kumar

M/s DHL

Offer Ref: F23_00167; Date 14.02.2023

Subject – Techno-Commercial Offer for Conveyor Automation

Dear Team,

We are pleased to submit our Techno-Commercial Offer in response to your RFQ

As you will note, we have studied your requirements in great depth and, along with the information collected during the workshops and calls with you, we have put together a detailed technical proposal laid out in various sections and sequenced to enable you to understand our proposed solution and to re-enforce our commitment to be your partner in this strategic initiative.

In subsequent sections, we have highlighted the capabilities and experiences of Falcon Autotech with sections on our Automation Technologies and references.

To conclude, I would like to add my personal commitment, on behalf of Falcon Autotech. As we move through the RFP process, please do not hesitate to contact me and my team. We will be pleased to assist you for any further information or clarifications that you might have.

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1. About Falcon

THE COMPANY

ABOUT US

Falcon Autotech is one of the leading Intra-logistics automation solution providers with a sharp focus on Parcel, Piece and Case Handling applications. We have a broad portfolio of Sortation Systems for Retail, E-Commerce and CEP (Courier, Express and Postal) industries and have supplied our systems across various countries globally.

OUR VISION

To create a global impact on the society by facilitating world's leading organizations run their supply chains efficiently by leveraging technology enabled products and solutions and, in the process, establish Falcon Autotech as one of the world's most respected and trusted brands in Intra-logistics warehouse automation space.

OUR JOURNEY

The team started out in 2004 solving special purpose automation problems for clients and later pivoted to building a standard technology stack spanning across Hardware, Firmware and Software to tackle bigger Supply Chain problems. Over the last few years, the organization has made rapid strides and has carved out a niche in some of the world's most cutting-edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.

OUR CORE COMPETENCIES

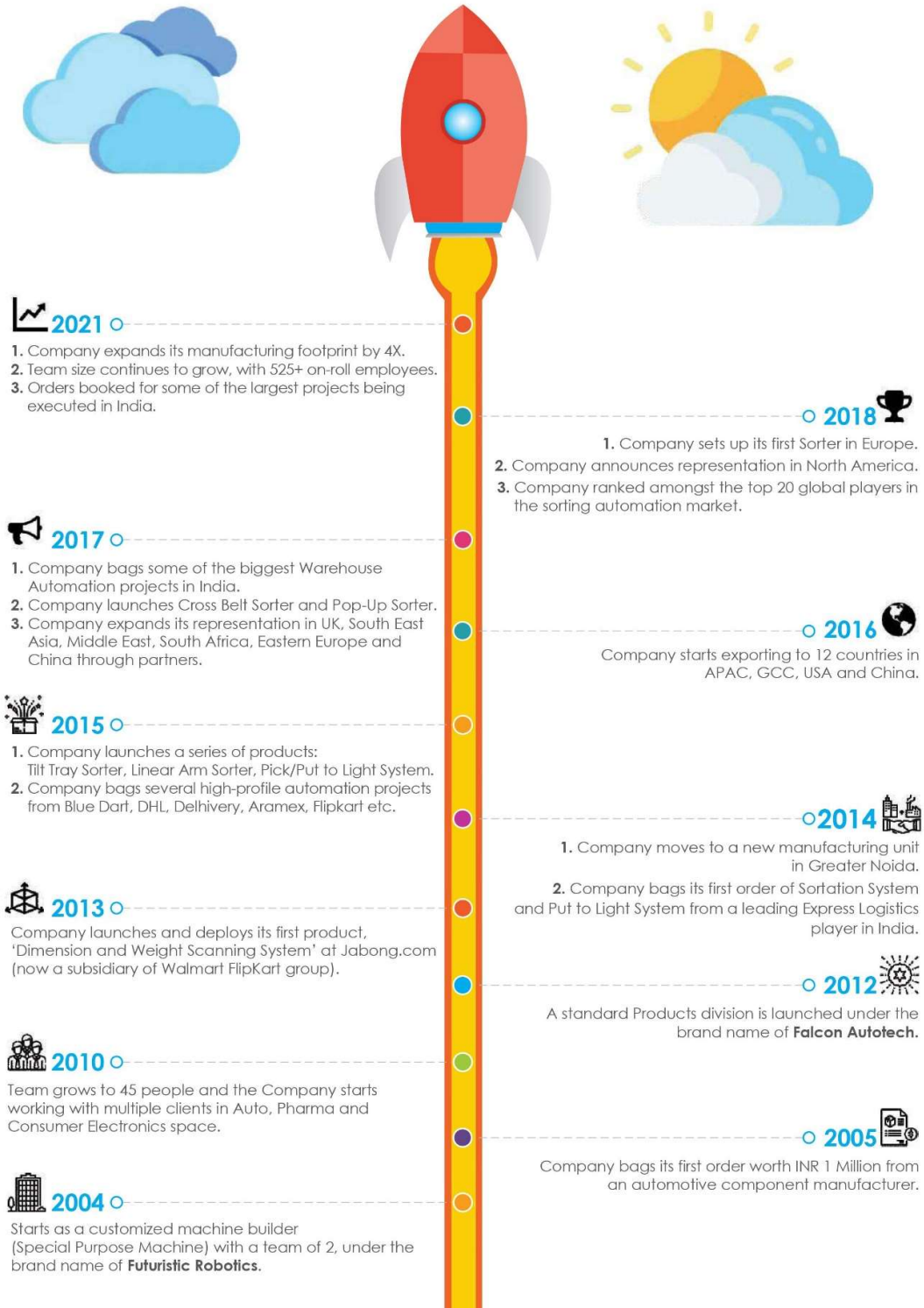
Organizations are built and defined by the people they are able to attract, cultivate and retain and the talent management philosophy at Falcon Autotech could not agree more. At, Falcon Autotech we take a lot of pride in being the preferred career choice for some of the finest talent in the country across multiple domains such as:

- ❖ Supply Chain Solution Designing and Consultation
- ❖ Hardware Design
- ❖ Electrical Design
- ❖ Electronics Design
- ❖ Software Development
- ❖ Manufacturing

Giving us an unprecedented ability to harness our true potential generated by culmination of Supply Chain Know-How, quality Hardware and flexible Software under one roof.



COMPANY HISTORY



KEY STATISTICS

**04**Product Lines

**525+**On-Roll Employees

**290+**Engineering Team

**400000+ Sq Ft**Manufacturing Space

**20+**Tier A Clients

**15+**Countries with Live Installations

**1800+**Total Installed Systems Globally

**35**Representations in 4 Continents and 35 Countries

**1 Million+**Sorts Per Hour of combined Sorting Capacity
installed Globally

OUR PRODUCT PORTFOLIO

01

SORTATION SOLUTIONS



Cross Belt Sorter



Tilt Tray Sorter



Arm Sorter



Swivel Wheel Sorter



Sweep Sorter



Pusher Sorter



Pop Up Sorter

02

PICK/PUT TO LIGHT SYSTEMS



Packet Sortation



Multi-Line Order Consolidation



Order Picking



Store Order Fulfillment



Spare Parts Distribution



Kitting



Multi Order Pick Cart

03

DIMENSION & WEIGHT SCANNING SYSTEMS

STATIC DIMENSION & WEIGHT SCANNING SYSTEMS



Cubizon-R



Cubizon-R THRU



Cubizon-R Cross



Cubizon-R ECO

DYNAMIC CONVEYOR BASED SYSTEMS

OUR 8 MODELS



OUR PRODUCT PORTFOLIO

04

CONVEYOR AUTOMATION SOLUTIONS

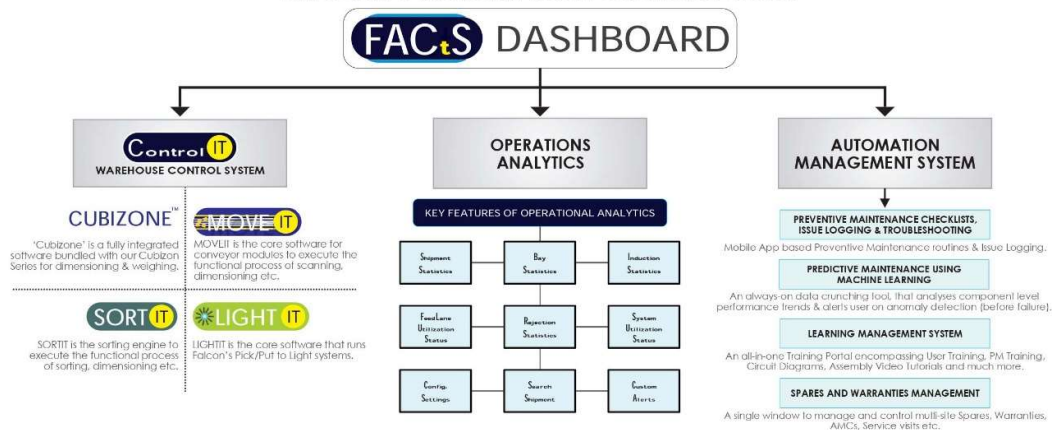
BELT CONVEYORS	ROLLER CONVEYORS	SPECIAL APPLICATION SOLUTIONS
❖ PVC BELT ◆ Straight Conveyor ◆ Spacing Conveyor ◆ Accumulation Conveyor ◆ Strip Belt Merge- 30/45/60 ◆ Solid Belt Merge- 45 ◆ Incline Belt Conveyor ❖ MODULAR BELT ◆ Straight Conveyor ◆ 90 Degree Belt Curve ◆ 180 Degree Belt Curve	◆ Straight Conveyor ◆ Gravity Roller Conveyor ◆ MDR Accumulation ◆ Angled Roller Conveyor ◆ Curved Conveyor ◆ Spur Merge Conveyor ◆ Swivel Wheel Diverter ◆ Pop Up Belt Diverter ◆ Pusher Unit	◆ Aligning Conveyor ◆ One to Two Split Conveyor ◆ Two to One Merge Module ◆ Bulk Distribution-ARB ◆ Bulk Distribution-Telescopic ◆ Bulk Distribution-Electric Arm ◆ Telescopic Belt Conveyor



05

SOFTWARE SUITE

FALCON'S AUTOMATION CONTROL SUITE



— ADD-ON MODULES —



Real Time Video Coding

No-Read Barcodes have always been a challenge in Sorter Applications globally. With Falcon's RTVC feature, we can now easily decode such challenging parcels in real time while sitting remotely.



Deep Learning Image Classification

DLC is an artificial intelligence function that imitates the working of the human brain. It can identify objects such as packets on the carrier and classify them in different categories.



Cloud Sort App/Virtual PTL

It's an android based app used to perform secondary sorting in light assisted put-away operations. It can work as a stand-alone setup or can be integrated with any sorter.

INDUSTRY SOLUTIONS

As a leading player in the intra-logistics automation space, Falcon Autotech continuously strives to improve the operational efficiencies and accuracies for its clients through its domain knowledge and experience in addition to its wide range of products and solutions.

In order to be able to live up to the high expectations set forth by our clients, the team at Falcon Autotech realizes the importance of taking up selective applications in focused Industries and deliver world class projects in return.

FOCUS APPLICATIONS FOR FALCON AUTOTECH

- ❖ Piece Picking and Handling
- ❖ Case Picking and Handling
- ❖ Parcel Handling

FOCUS INDUSTRIES



E-COMMERCE RETAIL



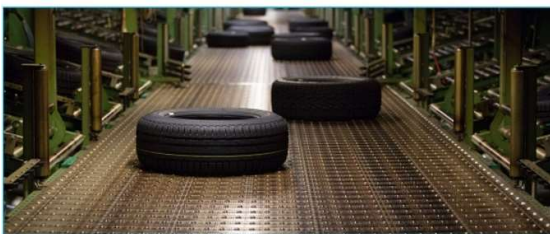
COURIER EXPRESS POSTAL



FASHION DISTRIBUTION



FFP (FOOD, FMCG, PHARMA) DISTRIBUTION



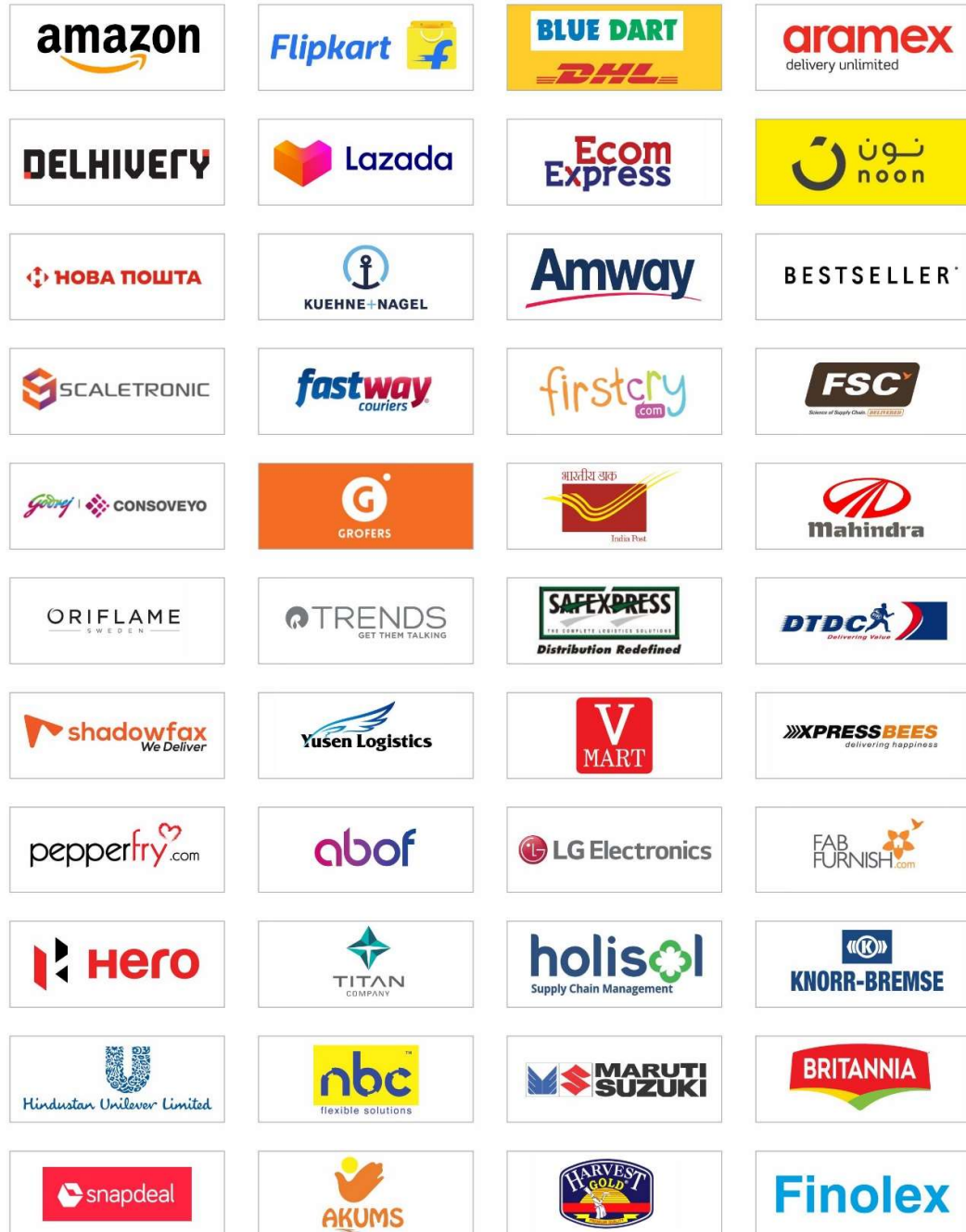
SPARE PARTS DISTRIBUTION



GROCERY DISTRIBUTION

SOME OF OUR GLOBAL CLIENTS

We believe in delivering high quality and standard worldwide. Here is a gist of our esteemed clients across various industry segments.



2. Handled Parcel Spectrum

2.1 Product size Conveyable on the entire system.

Max Length	mm	NA
Max Width	mm	600
Max Height	mm	NA
Max Weight	Kg	08
Min length	mm	NA
Min Width	mm	NA
Min Height	mm	NA
Min Weight	Kg	02

2.2 Parcels to be loaded on Conveyor/Sorter shall have the following characteristics:

1. Centre of Gravity of items must not move during conveyance or sorting.
2. Item must not have magnetic content; otherwise behaviour of parcel cannot be guaranteed.
3. Liquid or fragile material, to avoid breaking, spillage or leakage, such as wine bottles, metal cans of paint are designated as non- conveyable items.
4. Parcels shall be perfectly and safely packaged: protrusion or open surfaces are not allowed.
5. Plastic ropes shall be perfectly adherent to the surface of the package.
6. All items with the risk of being damaged during the transport on an automatic conveying/sorting system or damaging the system. They must be robust enough to avoid disintegration of container material and loose of contents in the conveying/sorting process.
7. Item packaging shall have enough grip to be handled on the belts during the acceleration and referencing phases.
8. Items shall not have slippery surfaces and must be able to withstand acceleration of the items on the belt during the start-stop phases (accelerations up to 0.5 g shall be assured without any sliding or tumbling of the items on the belt conveyor).
9. The parcels must have at least one flat and regular surface providing enough stability during conveyance.
10. All shapes are permitted except spherical, cylindrical, or alike unstable items & shapes.
11. All usual packaging materials are permitted (including paper, carton, plastics, plastic foil, rope, tape, textile, and wood)

2.3 Parcel not loadable on the system

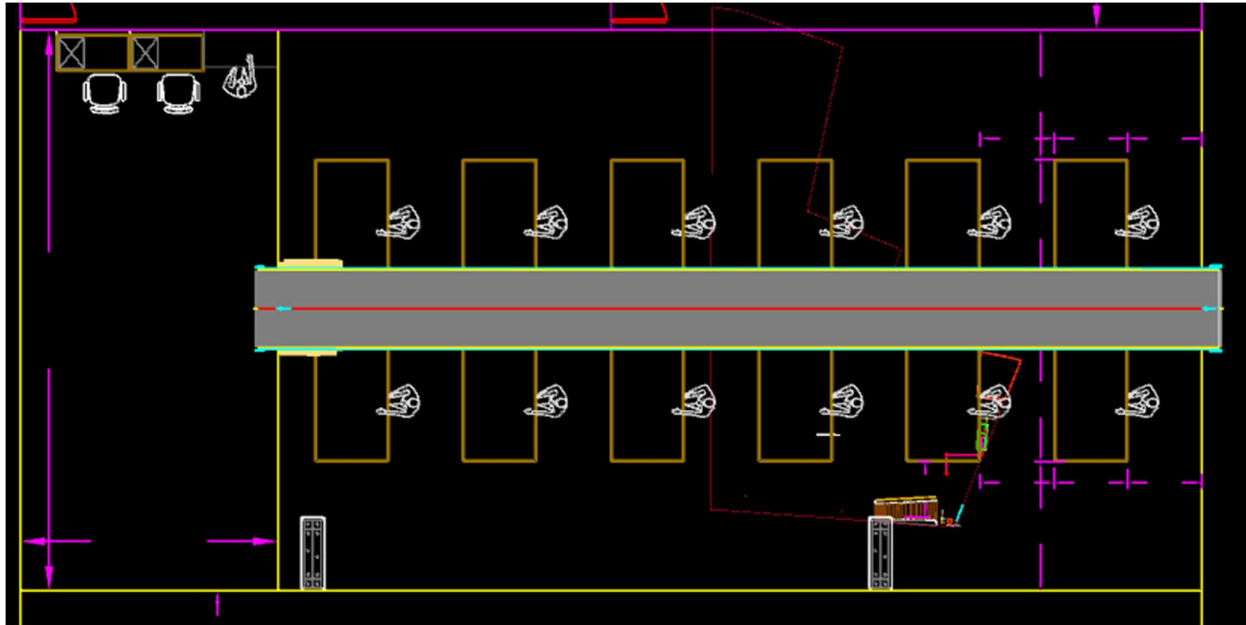
All products that are not within the range as described here, are considered non- conveyable products, and must be taken out of the main sorter flow by the operators.

1. Unstable items with a risk to roll or tumble on the sorting system, such as spherical or cylindrical items
2. Items that have a spherical or cylindrical shape.
3. Items that are packed in material that can damage the conveyors or the sorter.
4. Items that have sharp points (e.g. Nails) or sharp edges, that can damage the conveyors or the sorter.
5. Fragile parcels with contents not sufficiently secured
6. Items that have been classified as dangerous are designated.
7. Wet items are designated.
8. Items with anti-slip treatment.
9. Items with protruding parts.
10. Items with sharp edges.
11. Inadequately packed items that could be damaged during automatic transportation.
12. Electrostatically loaded items.
13. Loose parts on loads and load carriers, such as adhesive tape, stickers, slips of paper, straps, wrap foil etc. are designated as non- conveyable items.

3. Proposed System Description

3.1 Objective

The purpose of this proposal is to present the design, manufacturing, installation, commissioning, testing and acceptance testing of the Conveyor automation system to handle Corrugated boxes as per specifications shared by DHL.



3.2 Summary of the System

Solution is designed to transport the corrugated box containing Electronic devices along its length in specified direction.

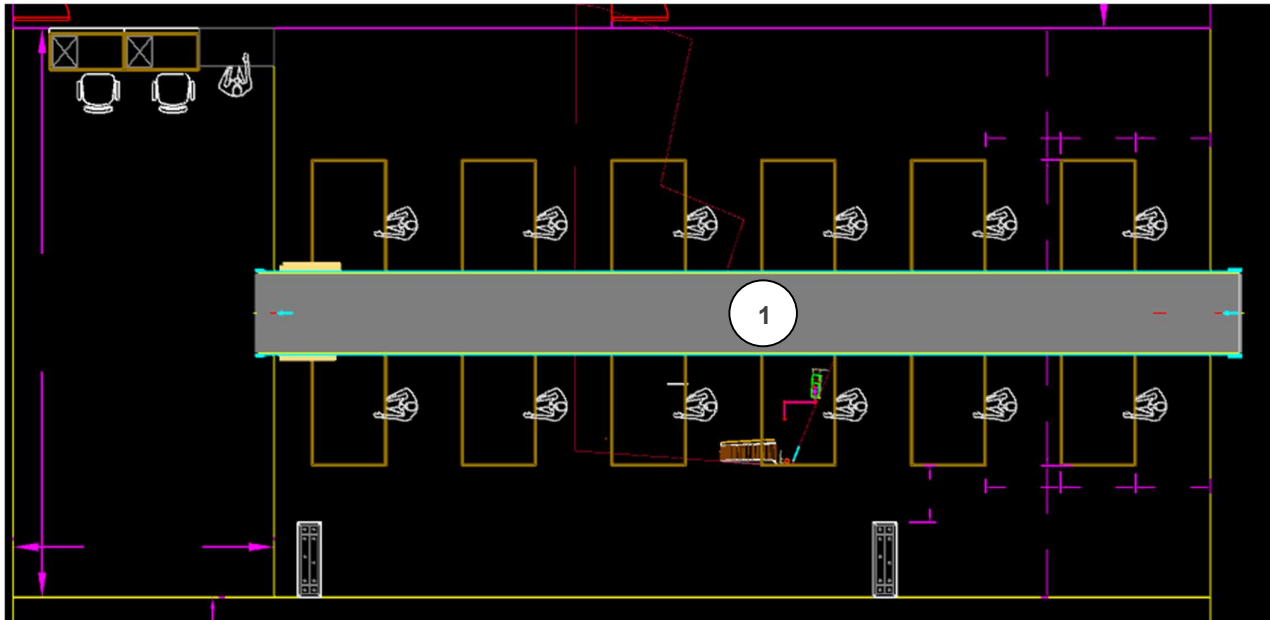
3.3 Main benefits of the proposed solution

The proposed solution has the following advantages:

1. Low occupancy of floor space in the building
2. FALCON's system can adapt with changing business requirements by adjusting its speed to match the operational throughput requirement, thereby leading to Power Savings and reduced system Wear & Tear

4. Layout View

4.1



Legend

1 – Belt Conveyor

5. Proposed System Capacity

5.1 System Capacity

Activity	Max Speed
Belt Conveyor	2m/s

6. System Components

6.1 PVC Belt Conveyor

Falcon's Belt Conveyors are modular & robust in design. MS profile is used to build conveyor frame.

Some Salient Features of Falcon's Belt conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
MOC of Belt	Type	PVC
Belt Thickness	mm	3
Belt Finish	Type	Matt Finish
Belt Joint Type	Type	Vulcanized, Endless Belts
Belt Make/Model No.	Make	FORBO/DERCO
Chassis	mm	MS, 3mm Thick, Powder Coated-75 Microns
Idler Roller Pitch on running side (Top)	mm	480
Idler Roller Pitch on return side (Bottom)	mm	1200
Idler Roller Specs	Spec	Dia.50.8, shaft-12 mm, bearing-6301
End Rollers	Spec	Dia 81.2 x 2.85 Thick, 30mm Shaft Dia, & Bearing Size-UCFH206D1, MOC-EN9, Zinc Plated -25 microns
Drive Rollers	Spec	OUTER Dia 178 PIPE Dia 168x5 Thick, 40mm Shaft Dia, & Bearing Size-UCF208D1, MOC(SHAFT)-EN9, NEOPRENE RUBBER COATING
Tensioning Rollers	Spec	Dia 81.2 x 2.85 Thick, 25mm Shaft Dia, & Bearing Size-UCT205D1, MOC-MS, Zinc Plated -25 microns
Take Up Rollers	Spec	Dia 81.2 x 2.85 Thick, 25mm Shaft Dia, & Bearing Size-6305-LLU, MOC-MS, Zinc Plated -25 microns
Type of Drive	Type	Rubberized
Angle of wrap/Contact - Belt with Drive pulley	Deg	>180 deg

Motor Shaft	MOC	EN9 Shaft material,
Conveyor Under guarding	Spec	MOC-MS, 0.8 to 1mm thickness, Steel Stiffeners at every 600mm to avoid the sagging due to self-weight
Load capacity per unit length of Conveyor	kg/m	50
Type of Guides	Type	Steel guides at 50mm/400mm height basis the position of conveyor on both sides from Conveyor Belt Surface.
Drive Power Rating	kW	Motor Rating depends on Length of the Conveyor
Type of Motor	Type	AC Geared Motor
Type of Drive System	Type	Direct Shaft Mounted Motor
Type of Drive	Type	Torque Arm type
Gear Motor	make	Sew
Drive	make	Lenze/Siemens/Omron/Allen Bradley
Insulation Class	Type	Yes
Conveyor Speed	m/s	Variable speed to meet TPH requirements
Type of Mounts	Type	Fixed Heights Legs with Grouting Provisions

1. Proposed System Technical Details/BOM

Mechanics		
Qty.	Description	Value
1	Conveyor Automation 01 x Powered Belt conveyor: L-14.9m	Width- 1200 mm Ht.800 mm
Controls		
1	Controls and Wiring Switches Drive panels Internal & Field Wiring	Included

2. Key Component Makes

Component	Make
VFD	Siemens/Omron/Lenze/AB
Switchgear	Schneider
Geared Motors	SEW
Belt Make	Movex/forbo
Bearings	NTN/SKF

3. Commercial Terms

3.1 Price Sheet

Sr. No	Component	Total (INR)
1	Conveyors	₹ 8,49,321
2	Toolkit	₹ 39,600
3	Installation & Commissioning	₹ 1,04,004
4	Packaging & Forwarding	₹ 1,17,778
5	Project Management and Engineering Charges	₹ 10,107
Total		₹ 10,20,810

The Total Price is to be understood.

- Freight charges: Included
- GST will be charged extra @ 18%
- Packing and Forwarding: Included
- Installation and Commissioning: Included.
- Price is valid for 60 days from the date of proposal

Our Price does not include taxes, and duties, import duties, charges, fees, taxes etc. charged by authorities outside India for deliveries and services. Same shall be borne & paid by client itself.

*Material Unloading & Laydown area is in customer scope

3.2 Payment Terms

The Supplier Quotation is based on the following payment schedule:

Stage 1: 50% Advance along with PO.

Stage 2: 50% along with 100% Taxes prior to dispatch..

4. Warranty Period

Falcons offered System comes with a standard warranty of 1 year (Starts from the date of beneficiary use) and post completion of warranty, customer can opt for Extended warranty. Warranty covers the following support:

- 24 X 7 Telephonic, Email and Remote Service Support when required
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in Exclusion Clause)

The warranty does not apply to replacement or repair of:

- Normal wear and tear.
- Consumables.
- Faulty articles continued:
 - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
 - Negligence or abnormal use of equipment.
 - Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
 - A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that supplied by Falcon.
- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.

5. Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation.

All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

- Servers
- Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
- Electrical power supply and wiring to the main control cabinets
- UPS for Controls and Drives
- Electrical Power for Installation
- Network Cabling up to the Main Server Rack
- Intermediate wiring to parts which are to be supplied by the Purchaser/others
- Emergency/Uninterruptable power supply
- Fire-alarm and fire protection devices
- Traffic and route markings
- Laydown Area
- Ram protection devices
- Cat walks, bridges, maintenance aisles and platforms- If not specified
- Assembly tools like forklifts and hoisting machines
- All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
- Any Kind of Civil work
- Any adjustment of the Supplier's scope of supply to local rules and regulations
- X-Ray machines
- Roller cages/ Pallets
- Simulation and 3D animation of the sorter system
- Interface with other equipment not specified in this offer.
- Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
- The supply and installation of fencing around the different corridors.
- Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.